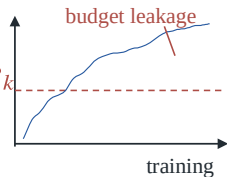


Penalty view

Costs weight the actor

$$A_r - \sum_k \lambda_k A_{c_k}$$

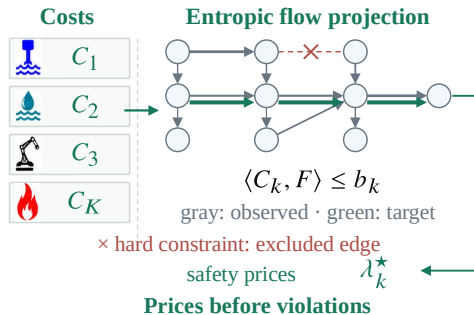


Fixed price can leave a budget violated.

(schematic)

SafeOT

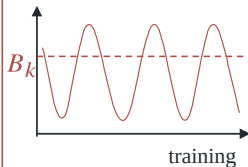
One flow prices all budgets



Constraint view

Budgets bound costs

$$J_{c_k}(\pi) \leq B_k$$



Lagrangian feedback can lag and oscillate.

(schematic)

$\lambda_k^* \rightarrow \text{clip} + \text{EMA} \rightarrow$

$$\lambda_k = \bar{\lambda}_k^* + \beta_k$$

observed-cost feedback

actor

PPO/TRPO

π_{t+1}

One price rule

Fixed penalty

$c = 0, \beta_k$ fixed

exact settings

Lagrangian

$c = 0, \beta_k$ adaptive

SafeOT

finite ε

LP

$\varepsilon \rightarrow 0$

conditional limits

Active-set

$\varepsilon \rightarrow \infty$, saturated clip